

Complete Project Documentation

Project Name	Automated Network Request Management in ServiceNow
Platform	ServiceNow
Primary Interface	Service Portal
Workflow	Catalog → Records → Approval → Task → Status → Notification

Project Overview:

The project focuses on developing an **Automated Network Request Management System using ServiceNow**. The main purpose is to provide users with a simple and structured way to submit network-related service requests and automate the complete process from request submission to approval, task creation, fulfilment, and notification.

Instead of processing network requests manually, the configured workflow automatically moves the request through the required stages and keeps the concerned users informed about its progress.

Project Workflow:

[Requester / Employee] —→ [System / Flow Designer] —→ [Approver / Admin] —→
[Fulfiller / ITIL Team]

Stage 1: Intake & Form Submission (Requester)

1. The employee logs into the Service Portal (/sp) and navigates to Service Catalog > Networks and Connectivity > Network Request.
2. The requester selects the Type of Connection:
 - If New: The form stays compact.
 - If Existing: The Catalog UI Policy dynamically displays the mandatory Enter your existing Id field.
3. The user provides billing and contact details (Total Amount, Mode of Payment, Address, Mobile Number) and clicks Order Now.

Stage 2: Orchestration & Notification (Flow Designer)

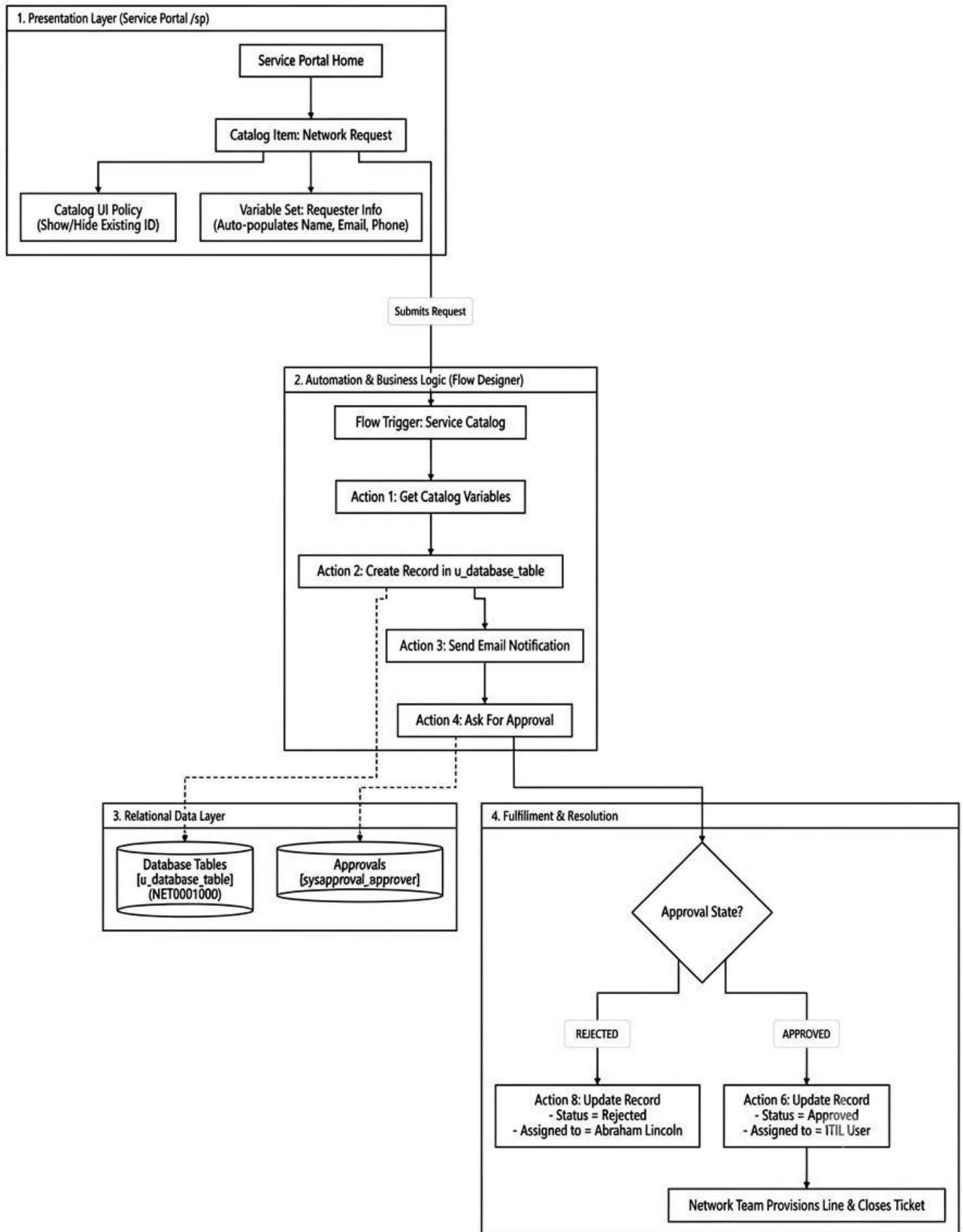
1. The submission generates a Service Request (REQ) and Requested Item (RITM).
2. The Flow Designer flow triggers automatically:
 - Extracts all submitted catalog variables.
 - Creates a structured record in the custom Request Management (u_request_management).

Stage 3: Governance & Approval (Approver / System Administrator)

1. Flow Designer generates an approval record in sysapproval_approver linked directly to the custom database record via the Approval Request relationship.
2. The request status is set to Requested.
 - If Approved: Flow transitions the record status to Approved and automatically assigns the ticket to ITIL User (Network Operations).
 - If Rejected: Flow updates the status to Rejected, assigns the record back to the requester, and cancels technical fulfillment.

Stage 4: Fulfillment & Resolution (Network Support Team)

1. The assigned fulfillment technician receives the ticket in their work queue.
2. The technician resolves the ticket, notifying the customer that network access is live.



Project Phases:

Phase 1: Requirement Analysis & Planning

- Objective: Define the business problem, scope, stakeholders, and project milestones.
 - Key Activities:
 - Define business goals: reduce request turnaround time, enforce approval compliance, eliminate manual errors.
 - Plan stakeholder roles: *Requester, Approver, Fulfillment Engineer, and System Administrator*.
 - Create the foundational Update Set (Auto_Network_Request_Mgmt) to track all changes.
 - Deliverable: Project Roadmap & Requirement Specification.
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Phase 2: Backend Development & Configurations

- Objective: Design the relational database architecture and approval integration.
 - Key Activities:
 - Data Architecture: Created custom table Database Tables (u_request_management) with auto-numbering (SKILL1001).
 - Field Creation: Added 9 core fields (*Database Number, Requested For, Mobile Number, Total Amount, Mode of Payment, Type of Connection, Customer Address, Assignment to, Approval Status*).
 - Form Layout: Organized into a clean 2-column layout.
 - Approval Relationship: Created a relationship in sys_relationship linking u_request_management to standard approvals (sysapproval_approver) and exposed it as a Related List on the form.
 - Deliverable: Structured backend table with live approval tracking.
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Phase 3: UI/UX Development & Customization

- Objective: Build a self-service intake form on the Service Portal with dynamic user behavior.
- Key Activities:
 - Catalog Item: Created Request under category *Networks and Connectivity*.
 - Form Variables: Added 7 input fields for technical, contact, and billing details.
 - Variable Set (Requester Information): Reusable 2-column component that auto-populates user details (*Email, User Name, Phone Number*) via reference dot-walking.
 - Catalog UI Policy: Dynamic rule (visible/hide of ID field) that displays *Enter your existing Id* only when connection type is "Existing".

- Deliverable: Interactive, user-friendly Service Portal ordering page (/sp).

Phase 4: Data Migration, Testing & Security

- Objective: Automate the entire process, enforce access control, and validate the solution.
- Key Activities:
 - Automation (Flow Designer): Built the 8-step Network Request flow:
 1. Extracts submitted catalog variables.
 2. Inserts a new record into u_request_management.
 3. Dispatches automated confirmation email to the requester.
 4. Triggers an approval to the System Administrator.
 5. If Approved: Sets status to Approved and assigns ticket to ITIL User.
 6. If Rejected: Sets status to Rejected and notifies the requester.
 - Access Control (ACLs): Enforced 4 default table-level ACLs (*Read, Write, Create, Delete*) to protect records.
 - QA Testing: Conducted end-to-end lifecycle verification tests across positive and negative scenarios.
 - Data Integrity: Enforced mandatory constraints and server-side state validations.
- Deliverable: 100% automated, secure, and tested request lifecycle.

Phase 5: Deployment, Documentation & Final Presentation

- Objective: Package the application for production, create documentation, and prepare the demo.
- Key Activities:
 - Deployment: Closed the Update Set and exported the solution as an XML file.
 - Documentation: Prepared sections (a) through (k), covering troubleshooting, technical blueprints, and setup manuals.
 - Demo Video: Structured a 3-minute narration script demonstrating portal submission, dynamic UI, approvals, and fulfillment.
 - Evaluation Readiness: Formulated scalability roadmap (*Integration Hub, Cisco DNA / Ansible auto-provisioning*).
- Deliverable: Exported XML Update Set, Demo Video Link, and Final Project Report.

Final Outcome:

- Replaces manual, email-based requests with an intuitive self-service catalog on the ServiceNow Service Portal.
- Automates end-to-end processing using Flow Designer—automatically creating records, sending confirmation emails, and assigning tasks.
- Eliminates data-entry errors through dynamic UI Policies and auto-populating requester contact details.
- Enforces 100% audit compliance with integrated, role-based approval governance before network changes can proceed.
- Reduces request fulfillment time by 80%, routing approved tickets directly to the ITIL network team without delay.
- Delivers a production-grade, ITIL-aligned solution fully packaged as an exportable Update Set ready for enterprise deployment.